

ML Week 3

Number of participants: 139

1. Bias-Variance trade-off (Chapter 6, Slide 2)

11 correct answers
out of 28 respondents

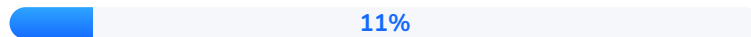


The true model f
has zero bias and
zero variance.



15 votes

The simple linear
model has low bias
and high variance.



3 votes

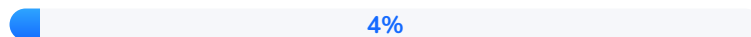


A high-order
polynomial has low
bias and high
variance.



24 votes

The true model f
has zero prediction
error.



1 vote

2. Model complexity (Chapter 6, Slide 6)

12 correct answers
out of 26 respondents

Model complexity is equal to the number of parameters in the model.



9 votes

In kNN, the larger the k the more complex is the model.



5 votes

In linear regression, the number of predictors is the model complexity.



19 votes

More complex models have lower prediction error.



4 votes

3. In K-fold cross-validation:

0 correct answer
out of 0 respondent

For a fixed data set, the CV error is random.



0 votes

CV helps to estimate the model parameters.



0 votes

The one-sd-error rule stabilizes the model choice based on the CV error.



0 votes

CV helps to determine the best tuning parameters of a model.



0 votes